

NLP Deep Learning Project – Part 2

Emotion Classification of Tweets: A Comparative Study of Transformer Models

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Abstract

This project analyzes Transformer-based models for multi-class emotion recognition. In Phase A, we compared BERTweet, Twitter-RoBERTa, and DeBERTa-v3. Statistical analysis revealed no significant difference in performance, with all models achieving F1 scores > 0.915 . Because of this, Twitter-RoBERTa was selected for one compression method due to its parameter efficiency (124M) and DeBERTa-V3 was selected for the other compression method due to runtime efficiency.

In Phase B, we implemented model compression to Twitter-RoBERTa via pruning two separate times. Once we pruned based on weight magnitude and separately we pruned an attention head. Second, we applied the distillation method to DeBERTa-V3 using DeBERTa-xsmall. Results show that the DeBERTa-V3 model is compressible: distillation to a DeBERTa-xsmall student achieved an F1 of 0.9125, getting very close to the teacher's performance. Pruning techniques yielded similar high performance with virtually no falloff from the non-pruned model's results on the validation dataset. We conclude that the transformers we tested for this task are robust to different architecture changes.

1. Introduction

We examine the task of taking a dataset of tweets from Twitter and classifying them into 6 emotions: 0: *sadness*, 1: *joy*, 2: *love*, 3: *anger*, 4: *fear*, 5: *surprise*. Emotion classification of tweets from Twitter is a unique challenge due to informal language and lack of unity between users in their sentiment expression. In this report, we compare transformer based models, some of which have “experience” with twitter form content, in terms of their classification performance. Afterwards, we perform methods of compression on the best performing models and compare with performance before compression.

2. Methodology

2.1 Data Preprocessing

To address the noisy nature of user-generated content, we implemented a preprocessing pipeline before tokenization. Our process includes the following steps: converting non-string values into empty strings, making all letter lowercase, opening up contractions to their full form (e.g. from ‘I’m’ to ‘i am’) - this is so a model which was trained on real vocabulary can “recognize” these words, removing duplicates of syllables which are more than 2 letters long to 2 letters (e.g. from ‘gooooooooood’ to ‘good’), removing punctuation and double spaces between words. We initially wanted to address also the problem of emojis in the tweets but we saw that no emojis exist in the data at all.

2.2 Phase A - Classification

2.2.1 Model Selection

Given that the task is sentiment analysis and emotion classification of tweets – we chose models that we saw perform well in that task according to the literature.

- **BERTweet** (vinai/bertweet-base): This model was introduced by Nguyen et al. (2020)^[1]. The model has the architecture of BERT_base but was pretrained based on RoBERTa method using 80GB pre-training dataset comprising of 850M tweets. **Table 1** compares different models on SemEval2017 Task4A^[2] (given a tweet – is it positive, negative or neutral) and shows BERTweet outperforms other models such as RoBERTa and XLM-R and other known benchmarks. In the article, they show that BERTweet also outperformed other models in SemEval2018 dataset Task 3A^[3] (detecting if a message is ironic or not). These tasks seem close enough to our task so this is a good model to try.

Model	AvgRec		F ₁ ^{NP}		Accuracy	
	soft	hard	soft	hard	soft	hard
Our results						
RoBERTa _{large}	72.5	72.2	72.0	71.8	70.7	71.3
XLM-R _{large}	71.7	71.7	71.1	70.9	70.7	70.6
RoBERTa _{base}	71.6	71.8	71.2	71.2	71.6	70.9
XLM-R _{base}	70.3	70.3	69.4	69.6	69.3	69.7
BERTweet	73.2	72.8	72.8	72.5	71.7	72.0
Cliche (2017)	68.1		68.5		65.8	
Baziotis et al. (2017)	68.1		67.7		65.1	

Table1 : Performance scores on the SemEval2017-Task4A test set, Nguyen et al

- **Twitter-RoBERTa** (cardiffnlp/twitter-roberta-base): This model was introduced by Barbieri et al. (2020)^[4]. The model (called RoB-RT in the article) is taken from a pre-trained RoBERTa base and then re-trained on

		Emoji	Emotion	Hate	Irony	Offensive	Sentiment	Stance	ALL
Val	SVM	25.0	63.8	73.1	63.4	72.7	68.4	67.9	62.0
	FastText	23.2	62.9	71.7	62.7	70.0	62.2	67.3	60.0
	BLSTM	19.4	62.6	72.1	60.6	72.1	61.9	63.4	58.9
	RoB-Bs	24.7±0.3 (24.3)	73.1±1.7 (74.9)	76.5±0.3 (76.6)	73.7±0.6 (73.7)	77.1±0.6 (77.6)	71.4±1.9 (72.7)	71.4±1.9 (73.9)	67.7
	RoB-RT	24.4±1.5 (26.2)	75.4±1.5 (77.0)	77.8±1.1 (79.6)	74.7±1.5 (75.6)	77.2±0.6 (77.7)	73.0±1.2 (74.2)	72.9±1.0 (75.2)	69.4
	RoB-Tw	23.4±1.1 (24.6)	67.6±0.9 (68.6)	74.3±2.0 (76.6)	70.0±0.3 (70.7)	76.1±0.6 (76.2)	70.5±1.0 (69.4)	68.3±2.4 (71.4)	65.4
Test	SVM	29.3	64.7	36.7	61.7	52.3	62.9	67.3	53.5
	FastText	25.8	65.2	50.6	63.1	73.4	62.9	65.4	58.1
	BLSTM	24.7	66.0	52.6	62.8	71.7	58.3	59.4	56.5
	RoB-Bs	30.9±0.2 (30.8)	76.1±0.5 (76.6)	46.6±2.5 (44.9)	59.7±5.0 (55.2)	79.5±0.7 (78.7)	71.3±1.1 (72.0)	68±0.8 (70.9)	61.3
	RoB-RT	31.4±0.4 (31.6)	78.5±1.2 (79.8)	52.3±0.2 (55.5)	61.7±0.6 (62.5)	80.5±1.4 (81.6)	72.6±0.4 (72.9)	69.3±1.1 (72.6)	65.2
	RoB-Tw	29.3±0.4 (29.5)	72.0±0.9 (71.7)	46.9±2.9 (45.1)	65.4±3.1 (65.1)	77.1±1.3 (78.6)	69.1±1.2 (69.3)	66.7±1.0 (67.9)	61.0
	Best	36.0*	-	65.1	70.5	82.9	68.5	71.0	-
Metric		M-F1	M-F1	M-F1	F ⁽ⁱ⁾	M-F1	M-Rec	AVG (F ^(a) , F ^(f))	TE

Table2 : Performance scores of emotion classification task on SemEval2018 dataset, Barbieri et al

Twitter data. The task the authors evaluated which is the closest to ours is emotion recognition by classification of tweets into 4 possible emotions: anger, joy, sadness and optimism. **Table 2** shows the results compared to other DL models (including RoB-Tw which is a RoBERTa model trained on Twitter data from scratch). We can see that for all tasks, including emotion recognition, RoB-RT outperforms the other models. Therefore, we decided to include this model in our project.

- **DeBERTa-V3** (microsoft/deberta-v3-base): DeBERTa was mentioned in class as the best BERT. Additionally, this model was not pre-trained on tweets so we thought it could be interesting to compare the best BERT with models which have “experience” with twitter form content. DeBERTa-V3 model was introduced by He et al (2023)^[5]. The model is an upgrade on the original DeBERTa model and utilizes ELECTRA style pretraining. The authors show the model does very well on different NLP datasets and tasks. This includes SST-2 task of analyzing sentiment and classifying a sentence into positive and negative.

2.2.2 Experimental Setup

Hyperparameters & Training Configuration: We fine-tuned the models on the training dataset using the same baseline configuration for each of the models. This includes a low learning rate of 2e-5 (after experimentation with higher rates proved unstable and moved the model to “forget” the pretrained weights). Additionally, to enable finetuning, we implemented a linear warmup of 500 steps. We utilized a batch size of 32 and applied a weight decay of 0.01 to

regularize the model and mitigate overfitting. Finally, we finetuned each model for only 5 epochs to prevent overfitting.

Statistical Evaluation: To verify our findings, we conducted Paired Sample T-Tests on precision, recall, and F1 scores across the six emotion classes.

Evaluation Metrics Selection: We selected the F1-Score (Macro Average) as our primary evaluation metric rather than Accuracy due to the significant class imbalance in our dataset (e.g., Joy is about 33.5% and Surprise about 3.5% of the total train dataset). Accuracy can be misleading in such scenarios, as a model could achieve high scores by biasing predictions toward majority classes while ignoring minority ones. The F1-score, being the mean of a measure which incorporates both Precision and Recall, notices variations between the two, and thus ensures minority classes are treated with equal importance to majority classes.

Results of Classification: For the fine-tuning process, for each model we output a plot of per epoch loss based on train dataset and per epoch loss based on validation dataset. This helps to measure whether overfitting occurred. Additionally, we ran each finetuned model on the validation dataset and output classification report, confusion matrix (for all the emotions) and accuracy and f1 measures per epoch. This allows us to evaluate how the model considered each class and how it behaved in classification while taking the imbalance mentioned above into consideration. All results and plots can be seen in the attached Jupyter notebook we handed in.

3. Classification - Model Comparison Results

3.1 Classification Results & Statistical Significance

Table 3 shows our results for the performance metrics of each model. To our pleasant surprise, all three architectures achieved very good results, with less than a 0.1% difference in F1 scores between them:

Model	F1-Macro	Training Time	Model Size
Twitter-RoBERTa	0.9164	30.7 min	124M
BERTweet	0.9163	30.2 min	135M
DeBERTa-v3	0.9155	23.2 min	184M

Table 3: Comparative performance metrics (F1-Macro) for the three evaluated models

To compare these models, we performed Paired T-Tests (with $\alpha = 0.05$) between all possible pairs of the models. The results yielded p-values > 0.05 for all pairwise comparisons:

- **RoBERTa vs. BERTweet:** $p = 0.9745$ (No significant difference).
- **RoBERTa vs. DeBERTa:** $p = 0.8918$ (No significant difference).

Conclusion: All three models are statistically tied in terms of performance. Therefore, our "Best" Model selection can rely purely on efficiency rather than accuracy. So we chose for the next Phase for compression the best model in terms of size (Twitter-RoBERTa) and the best model in terms of training time (DeBERTa-v3).

3.2 Overfitting Analysis

The result of training loss and validation loss per epoch helped us determine whether overfitting occurred. The results for each model were eerily similar in this plot and so one example (Figure 1) should be enough to discuss how the models did (rest can be seen in notebook). We can see that after the second epoch training loss continued to drop marginally while validation loss did not change.

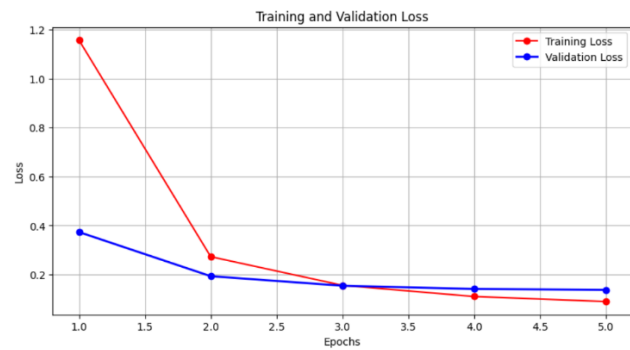


Figure 1 : Validation Loss vs. Training Loss BERTweet. The divergence after Epoch 3 indicates the onset of overfitting.

This is an important sign of the beginnings of overfitting, but because validation loss did not rise, we consider this ok for the few epochs we ran, as the model's performance on unseen data did not deteriorate. We conclude that 3-4 epochs is enough for this task and these models.

3.3 Error Analysis: The "Love" vs. "Joy" Ambiguity

Upon analyzing the Confusion Matrix, we observed that the percentage of correct predictions were high for distinct emotions like *Anger* and *Fear*. However, an interesting pattern of misclassification happened between Love and Joy. In different models the classification error between the two led to different biases. TwitterRoBERTa model was biased towards Love, DeBERTa-V3 towards Joy and BERTweet had a fairly even confusion between the two.

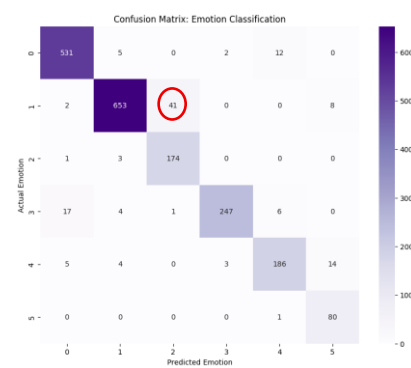


Figure 3: Confusion Matrix for RoBERTa. Note the semantic confusion between 'Joy' (1) and 'Love' (2), stemming from class imbalance and linguistic overlap.

This confusion can possibly be explained by being driven by both semantic overlap between the emotions (as people in love generally also feel joy) and class imbalance in the training data, where Joy (33.5%) significantly outnumbers Love (8.1%). Consequently, ambiguous statements like "I love hamburgers" (which imply enjoyment) are biased towards the dominant Joy label.

3.4 Other Notes Regarding Comparison of Results

- **Runtime & Size of Model:** DeBERTa-v3 emerged as the fastest model, completing training in 23.2 minutes. However, Twitter-RoBERTa required the least amount of parameters, requiring only 124M parameters, which is more than 10M parameters less than the next smallest model.
- **Convergence:** The results show that all three models converged quickly with the results of the first epoch for all models gave an F1-Macro score of at least 0.86. This indicates that both the models with experience in Twitter (BERTweet/Twitter-RoBERTa) and the non-experienced but still considered advanced architecture (DeBERTa-V3) provide good initialization for this task.

4. Phase B - Model Compression

4.1. Introduction

Since all 3 models performed similarly well in terms of our chosen measure for classification (f1-score), we chose a method of compression for each of the two better models. Twitter-RoBERTa was chosen for its smaller size (and thus better memory efficiency), and DeBERTa-V3 was chosen for its optimality in terms of runtime. After running each of the compression methods elaborated below, we had each compressed model output the same plots and reports as the previous stage.

4.2. Methodology

We implemented the following strategies:

4.2.1. Twitter RoBERTa - Pruning

We applied two pruning techniques on Twitter-RoBERTa followed by 2 epochs of finetuning again to retrain the model after pruning:

- Magnitude Pruning (Unstructured): Zeroing around 10% of the least important weights across the layers. To prune around this amount we implemented threshold of 0.005 for weights magnitude.
- Head Pruning (Structured): Removing 1 attention head from each layer which contributes the least to the model's performance. The result is the removal of 12 out of the 144 heads of the model. This head was determined by locating the minimum of the maximum attention weight each head assigns, as we learned in class.

4.2.2. DeBERTa-V3 - Knowledge Distillation (KD)

We used the transfer knowledge method from the teacher model to a student:

- Teacher: The fine-tuned DeBERTa-V3 implemented in the classification stage above.
- Student: microsoft/deberta-v3-xsmall (chosen for its compact size).
- Method: We employed a distillation loss function combining the soft targets (Teacher's logits) and hard targets (Ground Truth). We set $\alpha = 0.5$, giving equal weight to both losses, and applied a Temperature scaling of 4 to smooth the probability distributions.

4.3. Compression Results and Comparison

We compared the effectiveness of the three compression strategies against the Baseline (Twitter-RoBERTa). The results are summarized below:

Method	F1-Macro	Difference from Baseline	Efficiency Gain
Baseline (RoBERTa)	0.9164	-	-
Magnitude Pruning	0.9070	-0.0094	Sparse Weights
Head Pruning	0.9047	-0.0117	Removed Head
Baseline (DeBERTa-v3)	0.9155	-	86M Backbone Parameters
Distillation (DeBERTa-v3-xs)	0.9125	-0.003	Architecture Change

			22M Backbone Parameter
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4.3.1. Pruning

Both pruning methods demonstrated high performance on the validation dataset. Remarkably, the difference in performance on the F1-score between the TwitterRoBERTa and the pruned results are around a percentage of difference. The results about the pruned head was surprising. One of the heads which was pruned had a score of 0.71 importance (Layer 2 - Head 7). Despite this, the dropoff from the baseline was minimal. Since both these results are above 90%, we can conclude that this is still a very good result and TwitterRoBERTa can be compressed and still give a satisfactory performance in emotion detection of tweets.

4.3.2. Distillation

Knowledge Distillation was successful, achieving an F1 score of 0.9125 which is very similar to the parent. Even though the student model (DeBERTa-v3-xsmall) is significantly smaller and faster than the teacher, it matched the baseline performance (-0.003 F1 drop). This dropoff is much smaller than the dropoff we noticed from pruning. This result suggests that for the twitter emotion task, we can still use a much smaller model, and it will still work fairly well.

5. Conclusion

This part of our project (Part B) provided an evaluation of Transformer architectures for emotion classifications of tweets. In the classification phase, statistical analysis revealed that Twitter-RoBERTa, BERTweet, and DeBERTa-v3 perform equally well ($F1 > 0.915$) with no statistical significance between them. Twitter-RoBERTa was selected as the baseline for compression using pruning method due to its favorable parameter efficiency. DeBERTa-V3 was selected as the baseline for compression using distillation due to its favorable runtime.

In the compression phase, we demonstrated that the models are compressible with minimal loss to performance. Magnitude Pruning and Head Pruning retained >99% of the performance, confirming over-parameterization and dense architecture which some of it may be redundant. Additionally, Knowledge Distillation proved to be a good method as well, successfully transferring knowledge to a compact DeBERTa-v3-xsmall student with an F1 score of 0.9125, despite architectural differences.

Final Recommendation: For the test dataset, we recommend full Twitter-RoBERTa model as it brought the highest F1-macro score, which we defined as the measure with which we differentiate between the models.

Bibliography:

[1] Nguyen, D. Q., Vu, T., & Nguyen, A. T. (2020, October). BERTweet: A pre-trained language model for English Tweets. In *Proceedings of the 2020 conference on empirical methods in natural language processing: system demonstrations* (pp. 9-14).

[2] <https://alt.qcri.org/semeval2017/index.php?id=tasks>

[3] <https://competitions.codalab.org/competitions/17468>

[4] Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa-Anke, L. T. (2020). Unified benchmark and comparative evaluation for tweet classification. *arXiv 2020. arXiv preprint arXiv:2010.12421*.

[5] He, P., Gao, J., & Chen, W. (2021). Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*.